Homework 5

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```r
library(ggplot2)
```

```r
homicides = read.csv('data/homicide-data.csv') |>
  dplyr::mutate(city_name = paste0(city, ", ",state))
```

Import data Find the city with the highest homicide and convert to sf

```r
homicide.city.count = homicides |>
  dplyr::group_by(city, state) |>
  dplyr::summarise(count = dplyr::n()) |>
  dplyr::ungroup() |>
  dplyr::arrange(-count)

head(homicide.city.count)
```

```
## # A tibble: 6 x 3
##   city         state count
##   <chr>        <chr> <int>
## 1 Chicago      IL     5535
## 2 Philadelphia PA     3037
## 3 Houston      TX     2942
## 4 Baltimore    MD     2827
## 5 Detroit      MI     2519
## 6 Los Angeles  CA     2257
```

```r
chicago.homicides = homicides |>
  dplyr::filter(city == 'Chicago',
                state == 'IL') |>
  sf::st_as_sf(coords = c("lon", "lat"),
               crs = 4269) ## crs same as tigris object

chicago.homicides
```

```
## Simple feature collection with 5535 features and 11 fields
## Geometry type: POINT
## Dimension:     XY
## Bounding box:  xmin: -87.84657 ymin: 41.64791 xmax: -87.52605 ymax: 42.02271
```

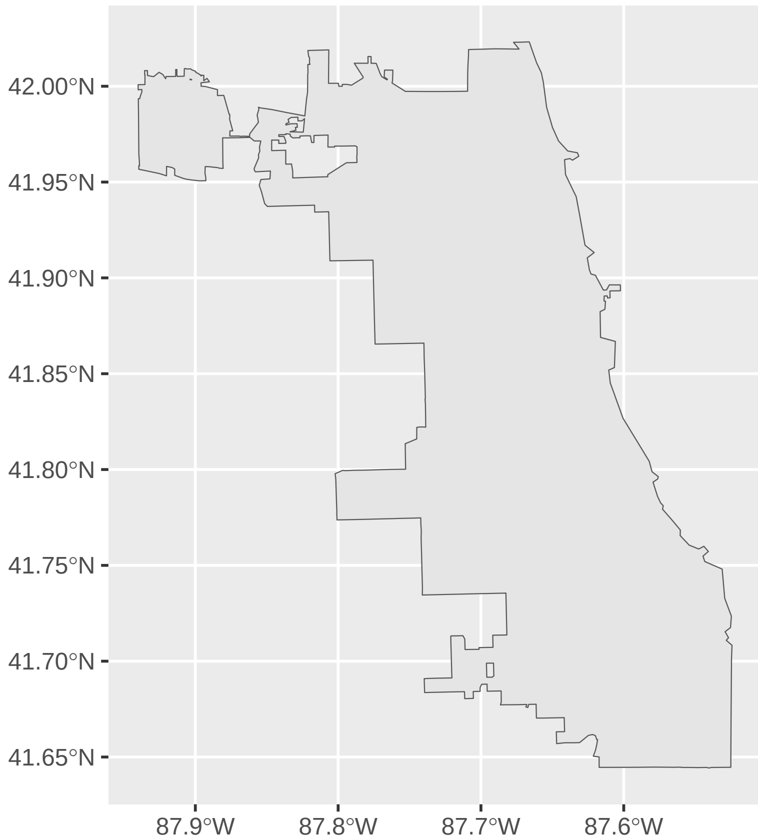
```
## Geodetic CRS:  NAD83
## First 10 features:
##           uid reported_date victim_last victim_first victim_race victim_age
## 1  Chi-003284      20070101      THOMAS         LEON       Black         36
## 2  Chi-003285      20070101       ALLEN        MARIO       Black         31
## 3  Chi-003286      20070101        HALL      CHALLAH       Black         21
## 4  Chi-003287      20070101      GUZMAN     CAMERINO    Hispanic         25
## 5  Chi-003288      20070102        ROOT        FRANK       White         34
## 6  Chi-003289      20070104       WHITE     MEREDITH       Black         37
## 7  Chi-003290      20070104    ABUNIMEH       RAMI M       White         31
## 8  Chi-003291      20070108        BELL       DENNIS       Black         38
## 9  Chi-003292      20070111     CAMBELL       RONNIE       Black         29
## 10 Chi-003293      20070112      DENTON        NAJJA       Black         31
##    victim_sex    city state            disposition   city_name
## 1        Male Chicago    IL      Closed by arrest Chicago, IL
## 2        Male Chicago    IL Closed without arrest Chicago, IL
## 3        Male Chicago    IL      Closed by arrest Chicago, IL
## 4        Male Chicago    IL        Open/No arrest Chicago, IL
## 5        Male Chicago    IL      Closed by arrest Chicago, IL
## 6        Male Chicago    IL        Open/No arrest Chicago, IL
## 7        Male Chicago    IL        Open/No arrest Chicago, IL
## 8        Male Chicago    IL        Open/No arrest Chicago, IL
## 9        Male Chicago    IL      Closed by arrest Chicago, IL
## 10       Male Chicago    IL        Open/No arrest Chicago, IL
##                      geometry
## 1   POINT (-87.67493 41.7668)
## 2  POINT (-87.74467 41.89165)
## 3  POINT (-87.58057 41.76514)
## 4  POINT (-87.71251 41.83797)
## 5  POINT (-87.62703 41.85043)
## 6  POINT (-87.73429 41.88064)
## 7  POINT (-87.72888 41.89809)
## 8  POINT (-87.62397 41.79015)
## 9  POINT (-87.66341 41.73853)
## 10 POINT (-87.62472 41.80106)
```

Import chicago vector

```
chicago.boundary = tigris::places(state = "IL", cb = T, class = "sf") |>
  dplyr::filter(NAME == "Chicago")
```

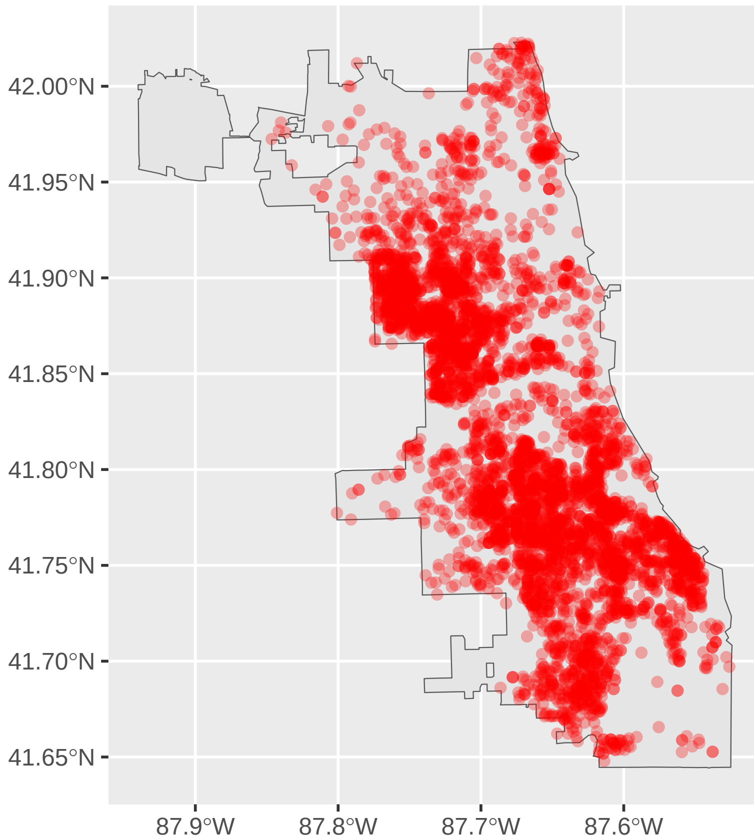
```
##   |                                                                              |
```

```
ggplot(data = chicago.boundary) +
  geom_sf()
```



Plot locations of homicides into chicago

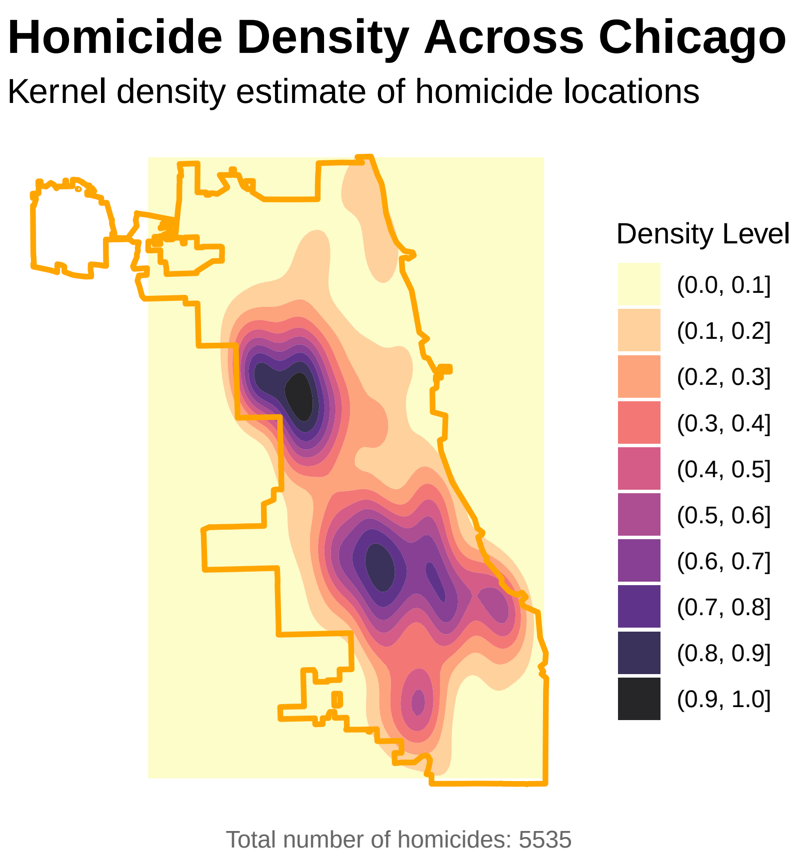
```
ggplot() +
  geom_sf(data = chicago.boundary) +
  geom_sf(data = chicago.homicides, colour = "red", alpha=0.3)
```



Trun map into heat map to show density

```
chicago.homicides.coords <- chicago.homicides |>
  sf::st_coordinates() |>
  as.data.frame() |>
  dplyr::bind_cols(chicago.homicides)


ggplot() +
  ## Density heat map
  geom_density2d_filled(
    data = chicago.homicides.coords,
    aes(x = X, y = Y, fill = after_stat(level)),
    alpha = 0.85,
    contour_var = "ndensity"
  ) +

  scale_fill_viridis_d(
    option = "magma",
    direction = -1,
    name = "Density Level"
  ) +
  ## Chicago BOundary
  geom_sf(data = chicago.boundary, color = "orange", size = 1, alpha = 0) +

  labs(
```

```
    title = "Homicide Density Across Chicago",
    subtitle = "Kernel density estimate of homicide locations",
    caption = paste0("Total number of homicides: ", nrow(chicago.homicides))
  ) +

  theme_minimal() +
  theme(
    panel.grid = element_blank(),
    axis.text = element_blank(),
    axis.title = element_blank(),
    legend.position = "right",
    plot.title = element_text(face = "bold", size = 18),
    plot.subtitle = element_text(size = 13),
    plot.caption = element_text(size = 9, color = "grey40")
  )
```



Interpretation: * Areas with highest homicide density in the area are in the darkest color contour * Each lighter contour shows areas with proportionally lower relative density to the darkest contour